The Fundamental Theorem of Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^3 (x^2 - 2x)\,dx$ using the Fundamental Theorem of Calculus.

$F(x) = \frac{x^3}{3} - x^2$. $F(3) - F(0) = (9 - 9) - (0) = 0$.

2 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_1^4 \sqrt{x}\,dx$.

$F(x) = \frac{2}{3}x^{3/2}$. $F(4) - F(1) = \frac{2}{3}(8) - \frac{2}{3}(1) = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$.

3 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^{\pi/2} \cos x\,dx$.

$F(x) = \sin x$. $F(\pi/2) - F(0) = 1 - 0 = 1$.

4 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^x t^2\,dt$. Find $g'(x)$ using FTC Part 1.

By FTC Part 1, $g'(x) = x^2$ (the integrand evaluated at x).

5 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_1^x \sin(t^2)\,dt$. Find $g'(2)$.

$g'(x) = \sin(x^2)$, so $g'(2) = \sin(4) \approx -0.757$.

6 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^{x^2} e^{-t}\,dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.

$g'(x) = e^{-x^2} \cdot \frac{d}{dx}(x^2) = 2xe^{-x^2}$.

7 [Foundational] • Target: Interpreting accumulation functions

A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau)\,d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.

$s(4) = \int_0^4 (3 - 2\tau)\,d\tau = [3\tau - \tau^2]_0^4 = (12 - 16) - 0 = -4$. This means the particle's net displacement from $t = 0$ to $t = 4$ is -4 m (it ends up 4 m to the left/behind its starting point).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Interpreting accumulation functions**

Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t)\,dt$, explain what feature of g's graph occurs where f crosses the x-axis, and where f has a local max.

Where f crosses the x-axis (changes sign), $g' = f$ changes sign, so g has a local extremum there. Where f has a local max, $g' = f$ has a local max, meaning g has an inflection point there (since $g'' = f'$ changes sign from $+$ to $-$).

9 **[Intermediate]** • **Target: Average value of a function**

Find the average value of $f(x) = x^2$ on $[0, 3]$.

Average $= \frac{1}{3-0}\int_0^3 x^2\,dx = \frac{1}{3}\left[\frac{x^3}{3}\right]_0^3 = \frac{1}{3}(9) = 3$.

10 **[Intermediate]** • **Target: Average value of a function**

The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \le t \le 24$. Find the average temperature over this period.

Average $= \frac{1}{24}\int_0^{24}\left[70 + 10\sin\left(\frac{\pi t}{12}\right)\right]dt$. The sine term integrates to 0 over a full period (24 hours $= 2$ full cycles), so the average $= 70°$.

11 **[Intermediate]** • **Target: Definite integrals with substitution-friendly integrands**

Evaluate $\int_0^1 2x(x^2+1)^3\,dx$ using $u = x^2 + 1$.

$du = 2x\,dx$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$. $\int_1^2 u^3\,du = \left[\frac{u^4}{4}\right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}$.

12 **[Intermediate]** • **Target: The Mean Value Theorem for integrals**

Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

The average value of $f(x) = x^2$ on $[0, 3]$ is 3 (found earlier). Set $f(c) = c^2 = 3$: $c = \sqrt{3} \approx 1.73$, which lies in $(0, 3)$.

13 **[Intermediate]** • **Target: Visualizing FTC Part 1 as an accumulation function**

The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2\,dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x.

Since $f(t) = t^2 \ge 0$ for all t, every strip added as x increases contributes non-negative area, so $g(x) = \int_0^x t^2\,dt$ is always increasing (or constant only at $x = 0$). By FTC Part 1, $g'(x)$ equals the height of the integrand curve at x, i.e. $g'(x) = f(x) = x^2$, matching the graph's height directly.

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14 **[Intermediate]** • **Target: A capstone FTC application**

Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \le t \le 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

Volume $= \int_0^6 (8 - t)\,dt = \left[8t - \frac{t^2}{2}\right]_0^6 = 48 - 18 = 30$ liters. Since $r(t) = 8 - t = 0$ at $t = 8$, the inflow rate drops to zero exactly when the tank presumably reaches capacity, consistent with the water level leveling off.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: FTC Part 1 with more complex bounds** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Let $g(x) = \int_3^{2x} \ln(t + 1)\,dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

$g'(x) = \ln(2x + 1) \cdot \frac{d}{dx}(2x) = 2\ln(2x + 1)$.

16 **[Challenging]** • **Target: A second average-value application** • **Cross-Lesson: 8.01 Area Between Curves**

A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \le t \le 8$. Find the average production rate over the shift.

Average $= \frac{1}{8}\int_0^8 (50 + 10t - t^2)\,dt = \frac{1}{8}\left[50t + 5t^2 - \frac{t^3}{3}\right]_0^8 = \frac{1}{8}\left(400 + 320 - \frac{512}{3}\right) = \frac{1}{8}\left(\frac{2160 - 512}{3}\right) = \frac{1648}{24} \approx 68.67$ units/hour.

17 **[Challenging]** • **Target: Interpreting accumulation functions** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Let $g(x) = \int_2^{-3x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

Final Answer: -432. $g'(x) = (-3x)^2 \cdot (-3) = (9x^2)(-3)$, so $g'(4) = (-12)^2 \cdot (-3) = 144(-3) = -432$.

18 **[Challenging]** • **Target: Average value of a function** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Let $g(x) = \int_0^{-5x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

Final Answer: -1125. $g'(x) = (-5x)^2 \cdot (-5) = (25x^2)(-5)$, so $g'(3) = (-15)^2 \cdot (-5) = 225(-5) = -1125$.

19 **[Challenging]** • **Target: Definite integrals with substitution-friendly integrands** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Let $g(x) = \int_{-3}^{4x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

Final Answer: 1024. $g'(x) = (4x)^2 \cdot (4) = (16x^2)(4)$, so $g'(4) = (16)^2 \cdot (4) = 256(4) = 1024$.

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20 [Challenging] • Target: The Mean Value Theorem for integrals • Cross-Lesson: 2.01 The Definition of the Derivative

Let $g(x) = \int_{-2}^{2x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

Final Answer: 72. $g'(x) = (2x)^2 \cdot (2) = (4x^2)(2)$, so $g'(3) = (6)^2 \cdot (2) = 36(2) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_{2}^{6} (6x^2 + 8x)dx$ using the Fundamental Theorem of Calculus.

a. 544 ✓

b. 546

c. 576

d. 542

$F(x) = 2x^3 + 4x^2$. $F(6) - F(2) = (576) - 32 = 544$.

22 [Foundational] • Target: FTC Part 1: accumulation functions

Evaluate $\int_{-1}^{0} (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. -2

b. 2

c. 3

d. 0 ✓

$F(x) = x^3 + x^2$. $F(0) - F(-1) = (0) = 0$.

23 [Foundational] • Target: Interpreting accumulation functions

Evaluate $\int_{0}^{2} (-6x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. -18

b. -22

c. -17

d. -20 ✓

$F(x) = -2x^3 - x^2$. $F(2) - F(0) = (-20) = -20$.

24 [Foundational] • Target: Average value of a function

Evaluate $\int_{-2}^{1} (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. 4

b. 8

c. 2

d. 6 ✓

$F(x) = x^3 + x^2$. $F(1) - F(-2) = (2) + 4 = 6$.

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25 [Foundational] • Target: Definite integrals with substitution-friendly integrands

Evaluate $\int_2^7 (6x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. 715 ✓

b. 735

c. 717

d. 713

$F(x) = 2x^3 + x^2$. $F(7) - F(2) = (735) - 20 = 715$.

26 [Foundational] • Target: The Mean Value Theorem for integrals

Evaluate $\int_{-2}^3 (6x^2)dx$ using the Fundamental Theorem of Calculus.

a. 70 ✓

b. 72

c. 54

d. 68

$F(x) = 2x^3$. $F(3) - F(-2) = (54) + 16 = 70$.

27 [Foundational] • Target: Visualizing FTC Part 1 as an accumulation function

Evaluate $\int_0^5 (6x^2 + 4x)dx$ using the Fundamental Theorem of Calculus.

a. 298

b. 303

c. 302

d. 300 ✓

$F(x) = 2x^3 + 2x^2$. $F(5) - F(0) = (300) = 300$.

28 [Foundational] • Target: A capstone FTC application

Evaluate $\int_2^5 (3x^2 - 6x)dx$ using the Fundamental Theorem of Calculus.

a. 50

b. 54 ✓

c. 56

d. 52

$F(x) = x^3 - 3x^2$. $F(5) - F(2) = (50) + 4 = 54$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: FTC Part 1 with more complex bounds

Let $g(x) = \int_{-3}^{-4x} t^2\,dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

a. -16

b. -256 ✓

c. 64

d. -260

$g'(x) = (-4x)^2 \cdot (-4) = (16x^2)(-4)$, so $g'(2) = (-8)^2 \cdot (-4) = 64(-4) = -256$.

30 [Intermediate] • Target: A second average-value application

Let $g(x) = \int_{0}^{3x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. 48

b. 435

c. 432 ✓

d. 144

$g'(x) = (3x)^2 \cdot (3) = (9x^2)(3)$, so $g'(4) = (12)^2 \cdot (3) = 144(3) = 432$.

31 [Intermediate] • Target: FTC Part 1 (Differentiating Integrals)

Let $g(x) = \int_{2}^{-4x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. -1028

b. 256

c. -1024 ✓

d. -64

$g'(x) = (-4x)^2 \cdot (-4) = (16x^2)(-4)$, so $g'(4) = (-16)^2 \cdot (-4) = 256(-4) = -1024$.

32 [Intermediate] • Target: FTC Part 2 (Evaluation Theorem)

Let $g(x) = \int_{3}^{2x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. 74

b. 36

c. 18

d. 72 ✓

$g'(x) = (2x)^2 \cdot (2) = (4x^2)(2)$, so $g'(3) = (6)^2 \cdot (2) = 36(2) = 72$.

33 [Intermediate] • Target: Integration by u-Substitution

Let $g(x) = \int_{1}^{-5x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. -2000 ✓

b. 400

c. -80

d. -2005

$g'(x) = (-5x)^2 \cdot (-5) = (25x^2)(-5)$, so $g'(4) = (-20)^2 \cdot (-5) = 400(-5) = -2000$.

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34 [Intermediate] • Target: Average Value of a Function

Evaluate $\int_0^1 2x(x^2+1)^2\,dx$ using $u = x^2+1$.

a. 1

b. $\frac{7}{3}$ ✓

c. $\frac{7}{2}$

d. $-\frac{7}{3}$

$du = 2x\,dx$. When $x=0$, $u=1$; when $x=1$, $u=2$. $\int_1^2 u^2\,du = \left[\frac{u^3}{3}\right]_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3} = \frac{7}{3}$.

35 [Intermediate] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^1 2x(x^2+4)^2\,dx$ using $u = x^2+4$.

a. $\frac{61}{2}$

b. 3

c. $\frac{61}{3}$ ✓

d. $-\frac{61}{3}$

$du = 2x\,dx$. When $x=0$, $u=4$; when $x=1$, $u=5$. $\int_4^5 u^2\,du = \left[\frac{u^3}{3}\right]_4^5 = \frac{125}{3} - \frac{64}{3} = \frac{61}{3} = \frac{61}{3}$.

36 [Intermediate] • Target: FTC Part 1: accumulation functions

Find the average value of $f(x) = x^2+4x+3$ on $[0, 6]$.

a. 27 ✓

b. 63

c. 28

d. 26

Average $= \frac{1}{6}\int_0^6 (x^2+4x+3)dx = \frac{1}{6}\left[\frac{x^3}{3} + 4\frac{x^2}{2} + 3x\right]_0^6 = 27$.

37 [Intermediate] • Target: Interpreting accumulation functions

Find the average value of $f(x) = -3x^2 - 4x$ on $[0, 5]$.

a. -95

b. -36

c. -35 ✓

d. -34

Average $= \frac{1}{5}\int_0^5 (-3x^2-4x)dx = \frac{1}{5}\left[-3\frac{x^3}{3} - 4\frac{x^2}{2}\right]_0^5 = -35$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: Average value of a function** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?

a. An inflection point, since g is always increasing when its integrand is positive.

b. Neither -- accumulation functions never have local extrema.

c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$. ✓

d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

By FTC Part 1, $g'(x) = f(x)$. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

39 **[Challenging]** • **Target: Definite integrals with substitution-friendly integrands** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?

a. Neither -- accumulation functions never have local extrema.

b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$. ✓

d. An inflection point, since g is always increasing when its integrand is positive.

By FTC Part 1, $g'(x) = f(x)$. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

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40 [Challenging] • Target: The Mean Value Theorem for integrals • Cross-Lesson: 2.01 The Definition of the Derivative

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 5$ and negative for $t > 5$. What feature does g have at $x = 5$?

a. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$. ✓

b. An inflection point, since g is always increasing when its integrand is positive.

c. A local minimum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.

d. Neither -- accumulation functions never have local extrema.

By FTC Part 1, $g'(x) = f(x)$. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.

41 [Challenging] • Target: FTC Part 1 (Differentiating Integrals) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find $\frac{d}{dx}[\int_0^{x^2} \cos(t)dt]$.

a. $2x\cos(x^2)$ ✓

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\sin(x^2)$

By FTC Part 1 with the Chain Rule: $\cos(u) \cdot \frac{du}{dx} = \cos(x^2) \cdot 2x = 2x\cos(x^2)$.

42 [Challenging] • Target: FTC Part 2 (Evaluation Theorem) • Cross-Lesson: 3.01 The Chain Rule

Evaluate $\int_0^{\pi} \sin x dx$.

a. 2 ✓

b. 0

c. 1

d. -2

Antiderivative is $[-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$.

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43 [Challenging] • Target: FTC Part 2 (Evaluation Theorem) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Evaluate $\int_0^1 e^{2x}dx$.

a. $\frac{e^2-1}{2}$ ✓

b. e^2-1

c. $\frac{e^2}{2}$

d. $2(e^2-1)$

$[\frac{1}{2}e^{2x}]_0^1 = \frac{1}{2}(e^2-e^0) = \frac{e^2-1}{2}$.

44 [Challenging] • Target: Integration by u-Substitution • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Evaluate the definite integral $\int_0^1 xe^{x^2}dx$.

a. $\frac{e-1}{2}$ ✓

b. $e-1$

c. $\frac{e}{2}$

d. $2(e-1)$

Let $u = x^2 \Rightarrow du = 2xdx$. When $x=0$, $u=0$; when $x=1$, $u=1$. Integral becomes $\frac{1}{2}\int_0^1 e^u du = \frac{1}{2}[e^1-e^0] = \frac{e-1}{2}$.

45 [Challenging] • Target: Average Value of a Function • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

What is the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 3]$?

a. 6 ✓

b. 9

c. 18

d. 3

$f_{\text{avg}} = \frac{1}{3-0}\int_0^3 (3x^2-2x)dx = \frac{1}{3}[x^3-x^2]_0^3 = \frac{1}{3}(27-9) = \frac{1}{3}(18) = 6$.

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